One-file mathematics and statistics

Patex example

A mean with its calculation

The observations are $x = (2, 4, 6, 8)$, so the sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{2+4+6+8}{4} = 5.$$

```r
x <- c(2, 4, 6, 8)
n <- length(x)
sample_mean <- mean(x)
sample_sd <- sd(x)
standard_error <- sample_sd / sqrt(n)
c(mean = sample_mean, standard_error = standard_error)
```

```
          mean standard_error
      5.000000       1.290994
```

The computed mean is 5, and its standard error is 1.291. The calculation and this sentence use the same R variables.

The Python and Node chunks below also run in this document build. The Node chunk uses a simple TypeScript type annotation supported by the Node 22 runtime; it is not a general TypeScript project build.

```python
python_values = [2, 4, 6, 8]
print("Python mean:", sum(python_values) / len(python_values))
```

```
Python mean: 5.0
```

```typescript
const typedMean: number = [2, 4, 6, 8].reduce((a, b) => a + b, 0) / 4;
console.log("Typed Node mean:", typedMean);
```

```
Typed Node mean: 5
```

Matrix notation

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, ordinary matrix multiplication gives

$$Av = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \end{pmatrix}.$$

The next plot is generated during the same PDF build.

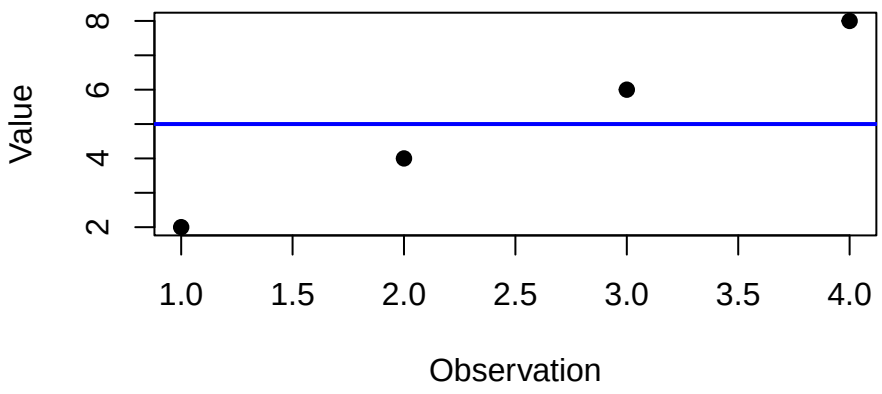


Figure 1: The four observations and their mean.

A citation can also live in this one source file through its YAML reference metadata (Fisher 1925).

Fisher, Ronald A. 1925. *Statistical Methods for Research Workers.*